Code No: 136AW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March - 2022
CRYPTOGRAPHY AND NETWORK SECURITY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Develop a model for network security and elaborate its components. [15]

2. Illustrate on 'substitution and transposition' techniques used for cryptography with example. [15]

3. Evaluate the key generation, encryption and decryption in Elgamal Cryptosystem. [15]

4.a) Describe RSA algorithm.
b) Perform encryption and decryption for the following using RSA algorithm. [7+8]
P=7, q=11, e=7, M=9.

5. Give the structure of SHA-512 compression function. Explain the structure of each round. Is MAN in the Middle attack possible on SHA-512? Explain. [15]

6.a) What is X.509 certificate? How it works and give the detailed format of X.509 certificate?
b) What is the job of key distribution center? [12+3]

7.a) What is the need for IEEE 802.11i wireless LAN security? Adapt a suitable solution for achieving security in IEEE 802.11i.
b) Interpret the benefit of HTTPS. [10+5]

8.a) Explain the general format of S/MIME.
b) Explain the basic combinations of security associations in IPSec. [7+8]

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